ENDG 319

Probability, Statistics and Machine Learning

Lecture 12: Exponential Distribution and the Central Limit Theorem

Waiting times • Memorylessness • Sampling distributions
Normal approximations • Continuity correction

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Learning outcomes for Lecture 12

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FROM WAITING TIMES TO APPROXIMATE NORMAL PROBABILITIES

1	**Model**	Recognize an exponential waiting time and interpret its rate.
2	**Calculate**	Use the PDF, CDF, tail probability, mean, and variance.
3	**Condition**	Apply memorylessness without confusing elapsed and extra time.
4	**Approximate**	Use the CLT for a sample mean, a sum, or a count.
5	**Adjust**	Apply continuity correction to integer-valued events.

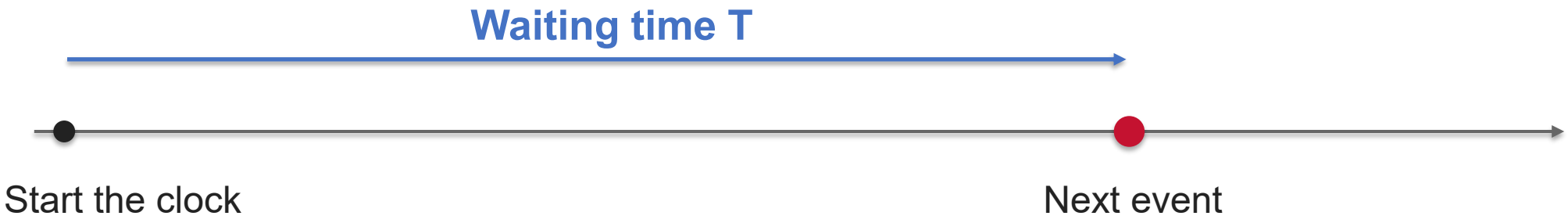
> Identify the random variable, check assumptions and units, then choose the probability rule.

The exponential model describes a waiting time

DEFINE WHAT IS RANDOM BEFORE WRITING A FORMULA

Let T be the time from a specified starting point until the next event.

$$T \sim \text{Exp}(\lambda), \qquad \lambda > 0$$



λ is a rate: its units are the reciprocal of the time unit.

A larger rate means a shorter characteristic waiting time—not a larger mean wait.

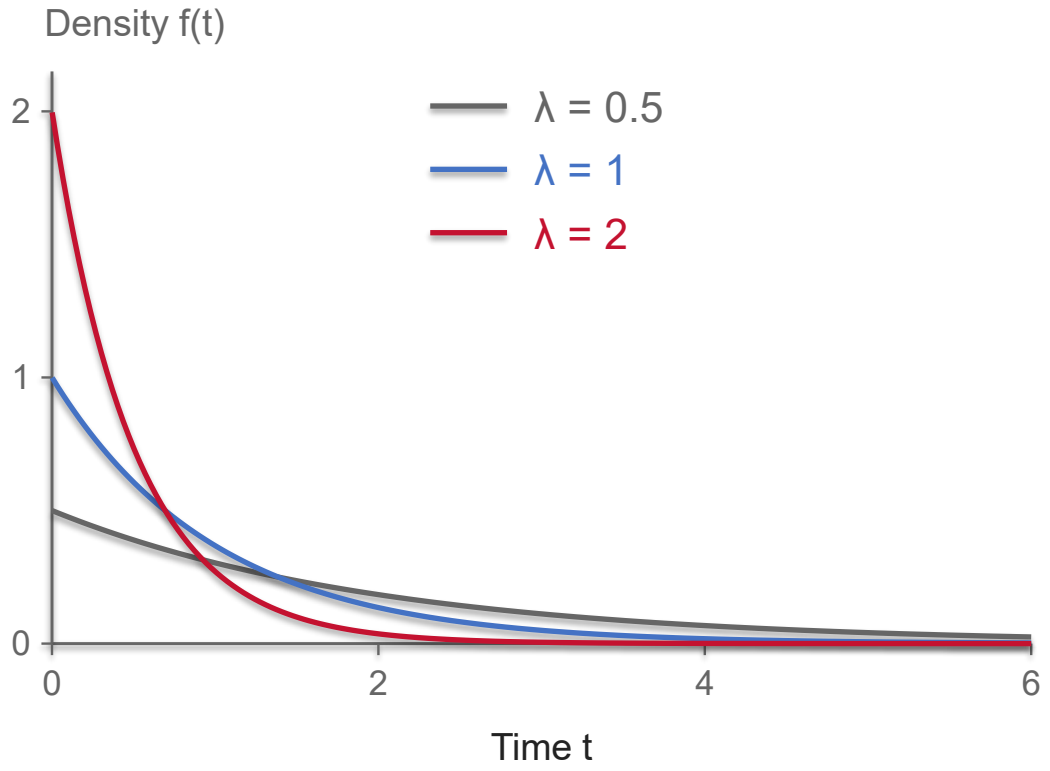
Exponential PDF: rate controls the time scale

NONNEGATIVE DENSITY WITH SUPPORT ON POSITIVE TIMES

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t > 0, \\ 0, & t \le 0. \end{cases}$$

Increasing λ concentrates density near the start of the time axis.

The curve height is a density.
Probabilities are areas.



The PDF integrates to one; its height need not be at most one.

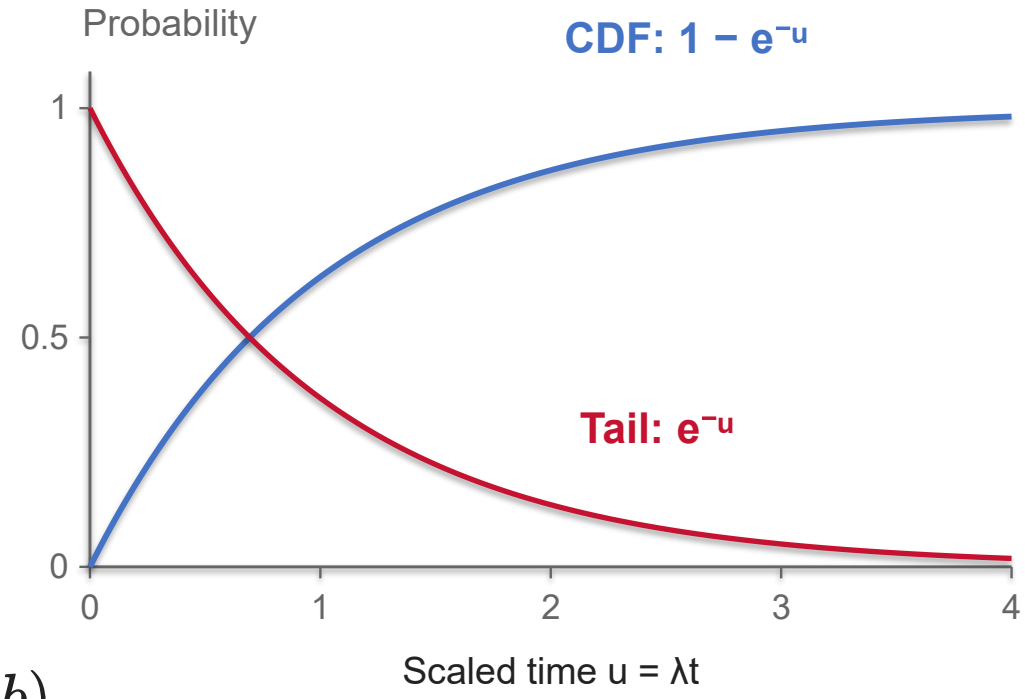
Exponential CDF and tail probability

“BY TIME t” AND “AFTER TIME t” ARE COMPLEMENTARY EVENTS

$$F_T(t) = \begin{cases} 1 - e^{-\lambda t}, & t > 0, \\ 0, & t \le 0. \end{cases}$$

By time t Use F(t) = 1 − exp(−λt).

After time t $P(T > t) = e^{-\lambda t}, \quad t \ge 0$



$$P(a < T \le b) = e^{-\lambda a} - e^{-\lambda b} \quad (0 \le a < b)$$

“More than t” uses the tail exp(−λt), not the CDF 1 − exp(−λt).

Exponential mean, variance, and standard deviation

THE RATE IS NOT THE MEAN WAITING TIME

MEAN

$$\mu_T = E[T] = \frac{1}{\lambda}$$

Units of time

VARIANCE

$$\sigma_T^2 = \mathrm{Var}(T) = \frac{1}{\lambda^2}$$

Squared time units

STANDARD DEVIATION

$$\sigma_T = \frac{1}{\lambda}$$

Units of time

Doubling λ halves both the mean and standard deviation; the variance becomes one quarter as large.

Check units: λ has units time^{-1}, while $1/\lambda$ has units of time.

One Poisson process: count events or measure a wait

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CONSTANT RATE λ • COUNT MEAN IN AN INTERVAL OF LENGTH t IS λt

COUNT IN A FIXED INTERVAL

N(t) is the number of events in time t.

$$N(t) \sim \text{Poisson}(\lambda t)$$

WAIT UNTIL THE NEXT EVENT

T is the elapsed time to the next event.

$$T \sim \text{Exp}(\lambda), \qquad \lambda > 0$$

$$\{T > t\} = \{N(t) = 0\}$$

$$P(T > t) = P\{N(t) = 0\} = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t}$$

No event before time t is exactly the same event as waiting longer than t.

Example 11: waiting for a particle emission

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A POISSON PROCESS WITH A MEAN RATE OF 15 PARTICLES PER MINUTE

A radioactive mass emits particles according to a Poisson process.
A clock is started at a specified point.

(1) Waiting-time probability

What is the probability that more than 5 seconds elapse before the next emission?

(2) Average waiting time

What is the mean waiting time until the next particle is emitted?

Before calculating: choose seconds or minutes and express both λ and t consistently.

Write the event and convert the time unit before substituting into $\exp(-\lambda t)$.

Example 11: convert units, then use the tail

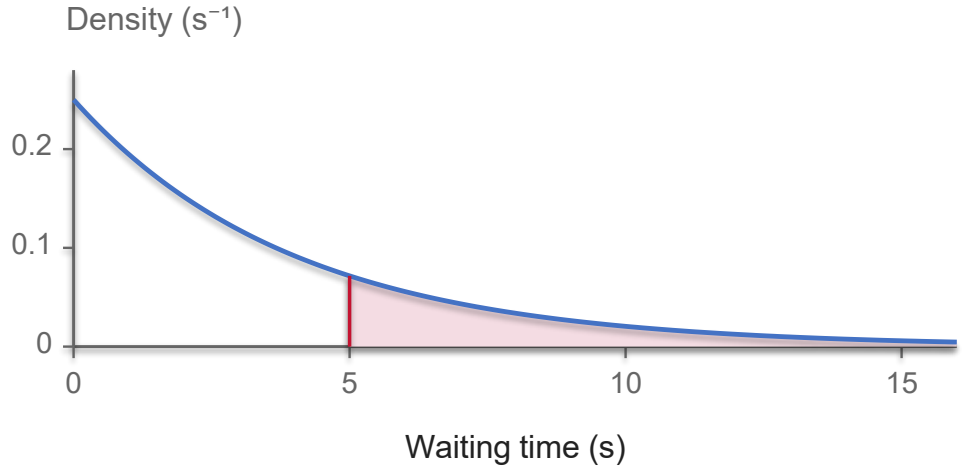
WORKED SOLUTION • KEEP THE EXPONENT DIMENSIONLESS

$$\lambda = 15\ \text{min}^{-1} = \frac{15}{60}\ \text{s}^{-1} = 0.25\ \text{s}^{-1}$$

$$P(T > 5) = e^{-(0.25)(5)} = e^{-1.25} \approx 0.2865$$

$$E[T] = \frac{1}{0.25} = 4\ \text{s}$$

About 28.65% of waits exceed 5 s.
The mean wait is 4 s.



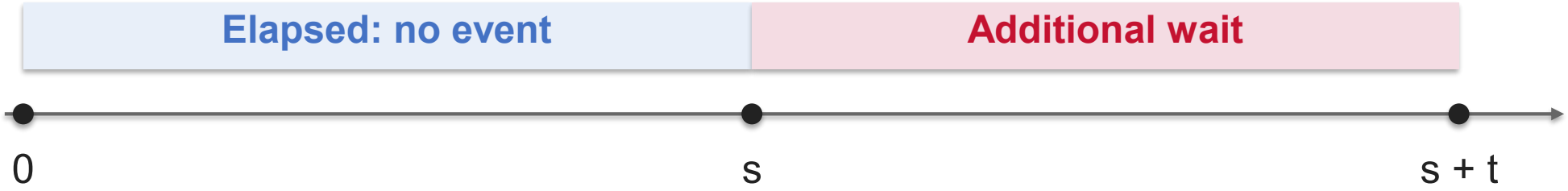
Using minutes gives the same exponent: 15 × (5/60) = 1.25.

Memorylessness: reset the waiting-time question

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CONDITION ON HAVING SURVIVED THE FIRST s UNITS

Already waited s units without an event.
Now ask about an additional t units.



$$P(T > s + t \mid T > s) = P(T > t), \quad s, t \geq 0$$

Only the additional duration t remains in the conditional tail probability.

Why the exponential model is memoryless

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NESTED EVENTS ALLOW THE CONDITIONAL-PROBABILITY RATIO

If T > s + t, then T > s. The numerator therefore reduces to P(T > s + t).

$$P(T > s + t \mid T > s) = \frac{P(T > s + t)}{P(T > s)}$$

$$\frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(T > t)$$

The factor exp(−λs) cancels. The result no longer depends on s.

This cancellation is not an independence assumption about the nested events.

Example 12: website hits and an additional minute

HITS FOLLOW A POISSON PROCESS AT 3 PER MINUTE

Let T be the time, in minutes, from starting the clock until the next website hit.

(1) No hit yet

What is the probability that more than one minute goes by without a hit?

(2) After two quiet minutes

Given that two minutes have passed without a hit, what is the probability of a hit in the next minute?

Part (2) asks about the interval from minute 2 to minute 3, given no earlier hit.

A hit in the next minute is the complement of waiting an additional full minute.

Example 12: condition, then take the complement

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WORKED SOLUTION • THE ELAPSED TWO MINUTES DO NOT ADD TO THE EXTRA WAIT

(1) More than one minute without a hit

$$P(T > 1) = e^{-3} \approx 0.0498$$

(2) A hit during the next minute, given T > 2

$$P(T \le 3 \mid T > 2) = 1 - P(T > 3 \mid T > 2)$$

$$= 1 - e^{-3(1)} \approx 0.9502$$

The answers are about 4.98% and 95.02%—a one-minute tail and its complement.

Exponential problems: four common mistakes

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CHECK THE MODEL, THE UNITS, THE EVENT, AND THE CONDITION

Mistake	Repair
Treat λ as a time.	λ is a rate; the mean time is 1/λ.
Use exp[−15 × 5] with minutes and seconds.	Convert units so λt is dimensionless.
Use the CDF for “wait longer than t.”	A longer wait is a right tail: exp(−λt).
Use s + t after conditioning on T > s.	The conditional remaining-time tail uses t.

Memorylessness is a property of the exponential model—not of every waiting time.

The CLT is about a distribution of sample means

A SINGLE SAMPLE MEAN IS ONE NUMBER; REPEATED SAMPLES GIVE A DISTRIBUTION

Sample 1: $X_1, \ldots, X_n$ → $\bar{x}_1$

Sample 2: $X_1, \ldots, X_n$ → $\bar{x}_2$

Sample 3: $X_1, \ldots, X_n$ → $\bar{x}_3$

Repeat the sampling procedure many times

The resulting mean values form the sampling distribution of $\bar{X}$.

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

The CLT does not make the original population or individual observations become normal.

The Central Limit Theorem: statement and conditions

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INDEPENDENT, IDENTICALLY DISTRIBUTED OBSERVATIONS WITH FINITE VARIANCE

Let $X_1, \ldots, X_n$ be independent and identically distributed (iid), with

$$E[X_i] = \mu, \qquad 0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$$

SAMPLE MEAN

$$\bar{X} \overset{\text{approx.}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

SAMPLE SUM

$$S_n \overset{\text{approx.}}{\sim} N(n\mu, n\sigma^2)$$

These normal laws are approximations when n is sufficiently large.

The CLT requires the stated sampling assumptions; “any population” is not unconditional.

Do not mix up an observation, a mean, and a sum

THE STANDARD DEVIATION MUST BELONG TO THE VARIABLE IN YOUR EVENT

Variable	Mean	Variance	Standard deviation
One observation X_i	μ	σ^2	σ
Sample mean $\bar{X}$	μ	σ^2/n	$\sigma/\sqrt{n}$
Sample sum S_n	$n\mu$	$n\sigma^2$	$\sigma\sqrt{n}$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \overset{\text{approx.}}{\sim} N(0,1)$$

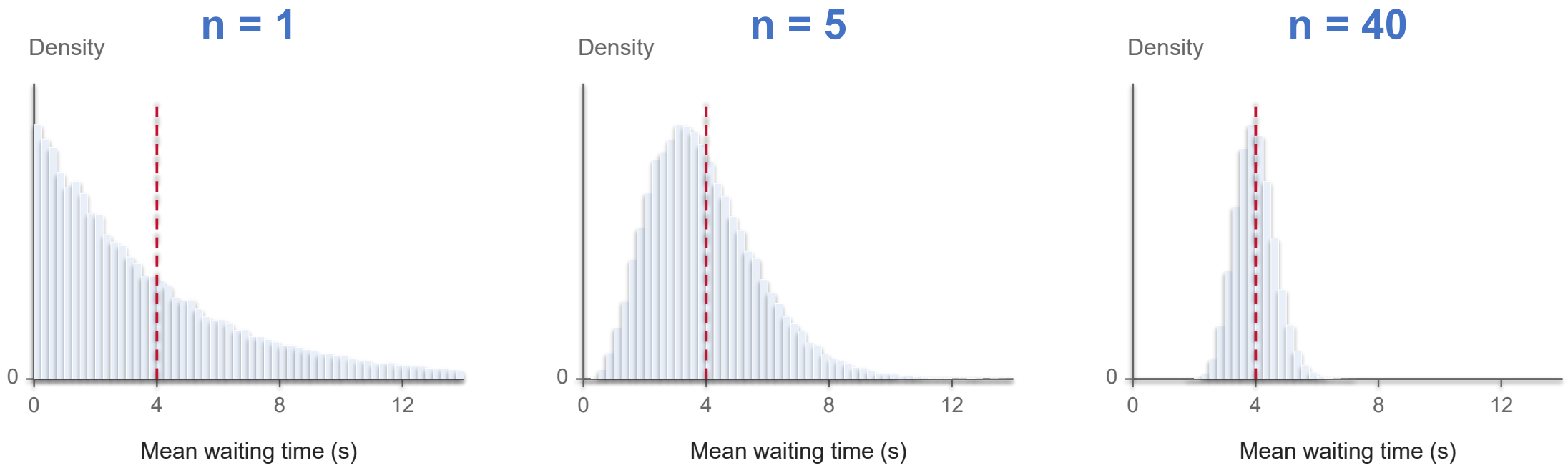
$$Z = \frac{S_n - n\mu}{\sigma\sqrt{n}} \overset{\text{approx.}}{\sim} N(0,1)$$

A mean divides the spread by $\sqrt{n}$; a sum multiplies it by $\sqrt{n}$.

A skewed population can produce nearly normal means

SIMULATION FROM EXAMPLE 11: INDEPENDENT EXPONENTIAL WAITS WITH MEAN 4 s

Common time axis; vertical density scales differ.



Strong right skew

Less skew, same mean

Narrower and more symmetric

Only the sampling distribution changes: individual waits remain exponential.

How large is “large enough” for the CLT?

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APPROXIMATION QUALITY DEPENDS ON THE UNDERLYING DISTRIBUTION

NEARLY SYMMETRIC POPULATION

A normal approximation may be useful with a relatively small n.

STRONGLY SKEWED POPULATION

A larger n may be needed.
Check the requested tail as well.

Source rule of thumb: n > 30

A working guideline—not a universal guarantee.

A large n does not repair selection bias or justify an unexamined independence assumption.

Normal approximation to a binomial count

CHECK BOTH EXPECTED SUCCESSES AND EXPECTED FAILURES

$$X \sim \text{Bin}(n, p) \qquad np > 10, \qquad n(1-p) > 10$$

$$X \overset{\text{approx.}}{\sim} N(np, np(1-p))$$

The sample proportion rescales the same count:

$$\hat{p} = \frac{X}{n} \overset{\text{approx.}}{\sim} N\left(p, \frac{p(1-p)}{n}\right)$$

The second normal parameter is the variance; the count SD is √[np(1 − p)].

Normal approximation to a Poisson count

SOURCE CORRECTION: THE NORMAL VARIANCE IS λ, NOT λ²

$$X \sim \text{Poisson}(\lambda)$$

Source working guideline: λ > 10

$$\mu_X = \lambda, \qquad \sigma_X^2 = \lambda, \qquad \sigma_X = \sqrt{\lambda}$$

$$X \overset{\text{approx.}}{\sim} N(\lambda, \lambda)$$

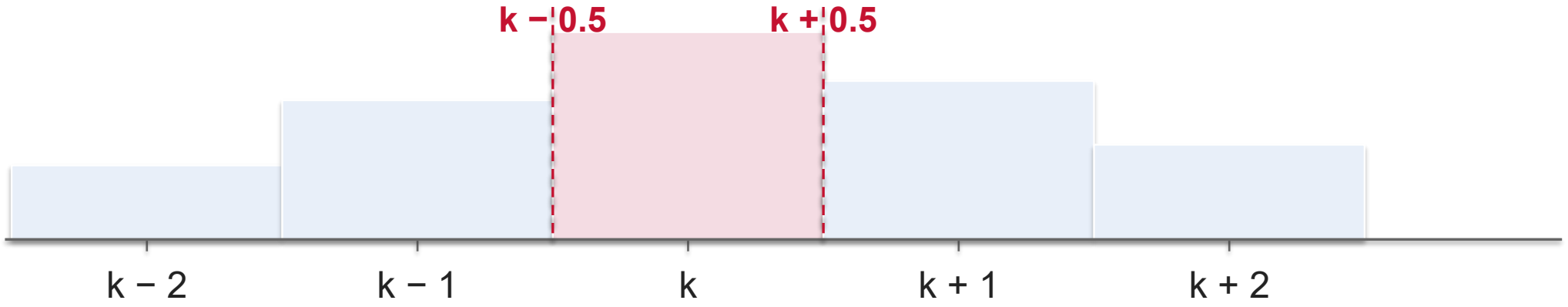
Here λ is the mean count in the specified interval, not a waiting-time rate.

Use √λ when standardizing a Poisson count, and check the integer endpoint correction.

Continuity correction: give each integer a width

A COUNT AT k CORRESPONDS TO A UNIT-WIDTH STRIP CENTRED ON k

Approximate a discrete probability mass with a continuous area spanning half a unit on either side of the integer.



$$P(X = k) \approx P(k - 0.5 < Y < k + 0.5)$$

Apply the ±0.5 adjustment to the count boundary before converting to a z-score.

Translate the inequality before correcting the boundary

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X IS INTEGER-VALUED; Y IS ITS CONTINUOUS NORMAL APPROXIMATION

Discrete event	Continuous event to approximate it
$X \le k$	$Y < k + 0.5$
$X < k$	$Y < k - 0.5$
$X \ge k$	$Y > k - 0.5$
$X > k$	$Y > k + 0.5$
$a \le X \le b$	$a - 0.5 < Y < b + 0.5$
$a < X < b$	$a + 0.5 < Y < b - 0.5$

Included integer endpoints keep their full strips; excluded endpoints lose their strips.

Example 13: total processing time for 65 parts

TWO INDEPENDENT MACHINE OPERATIONS ARE REQUIRED FOR EACH PART

Operation time per part	Mean (h)	Standard deviation (h)
Machine 1	0.40	0.10
Machine 2	0.45	0.15

For 65 parts, determine:

1. The distribution of total time on machine 1.
2. The distribution of total time on machine 2.
3. The probability that their combined total is between 50 and 55 hours.

Add aggregate machine-hours; do not calculate the elapsed time of a parallel schedule.

Example 13: define the sums and state the assumptions

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WORKED-SOLUTION ASSUMPTION: INDEPENDENT REPEATED PARTS

X_i = machine-1 time for part i.
Y_i = machine-2 time for part i.

$$S_1 = \sum_{i=1}^{65} X_i, \qquad S_2 = \sum_{i=1}^{65} Y_i, \qquad S = S_1 + S_2$$

To use the source's CLT approach:

Assume different parts have independent processing times, the stated moments apply to every part, and the two machine-time sequences are independent.

Under these assumptions, the moments are exact; normal shape remains approximate.

Example 13: total time on each machine

MULTIPLY EACH MEAN AND VARIANCE BY 65

MACHINE 1

$$E[S_1] = 65(0.40) = 26.00 \text{ h}$$

$$\text{Var}(S_1) = 65(0.10)^2 = 0.6500 \text{ h}^2$$

$$S_1 \overset{\text{approx.}}{\sim} N(26.00, 0.6500)$$

MACHINE 2

$$E[S_2] = 65(0.45) = 29.25 \text{ h}$$

$$\text{Var}(S_2) = 65(0.15)^2 = 1.4625 \text{ h}^2$$

$$S_2 \overset{\text{approx.}}{\sim} N(29.25, 1.4625)$$

Square the per-part standard deviation first; N(μ, σ²) takes variance as its second parameter.

Example 13: combine the two machine totals

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INDEPENDENT TOTALS: ADD MEANS AND ADD VARIANCES

$$E[S] = 26.00 + 29.25 = 55.25 \text{ h}$$

$$\text{Var}(S) = 0.6500 + 1.4625 = 2.1125 \text{ h}^2$$

$$S \overset{\text{approx.}}{\sim} N(55.25, 2.1125) \qquad \sigma_S = \sqrt{2.1125} \approx 1.4534 \text{ h}$$

The interval 50–55 h lies below the expected total, 55.25 h.

> Sum variances—not standard deviations—before taking the square root.

Example 13: standardize the continuous-time interval

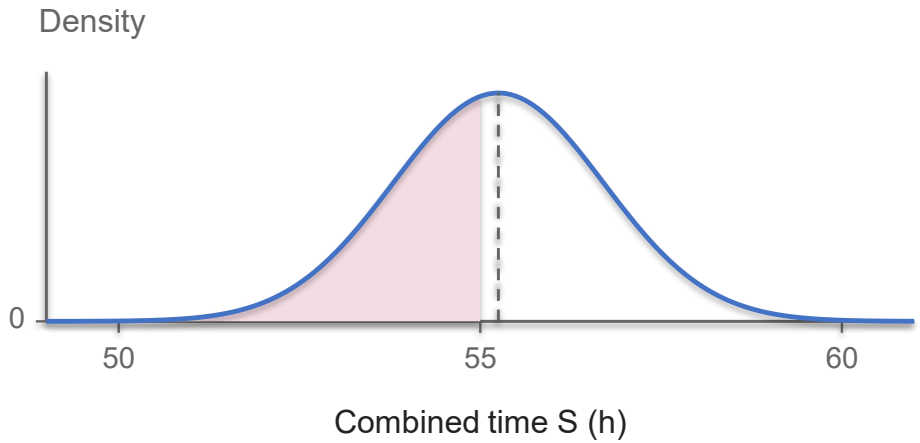
NO CONTINUITY CORRECTION: S IS A CONTINUOUS TIME

$$z_L = \frac{50 - 55.25}{\sqrt{2.1125}} \approx -3.6121 \qquad z_U = \frac{55 - 55.25}{\sqrt{2.1125}} \approx -0.1720$$

$$P(50 < S < 55) \approx \Phi(-0.1720) - \Phi(-3.6121)$$

$$\approx 0.4316 \quad (\text{about } 43.2\%)$$

Using unrounded z-scores.
A two-decimal z-table gives ≈ 0.4323.



Check plausibility: because 55 h is below the mean, the probability should be below 0.5.

Example 14: heads in 100 fair-coin tosses

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USE A NORMAL CURVE TO APPROXIMATE AN INTEGER-VALUED EVENT

A fair coin is tossed 100 times. Find an approximation to the probability of 45 to 55 heads, inclusive.

$$X \sim \text{Bin}(100, 0.5)$$

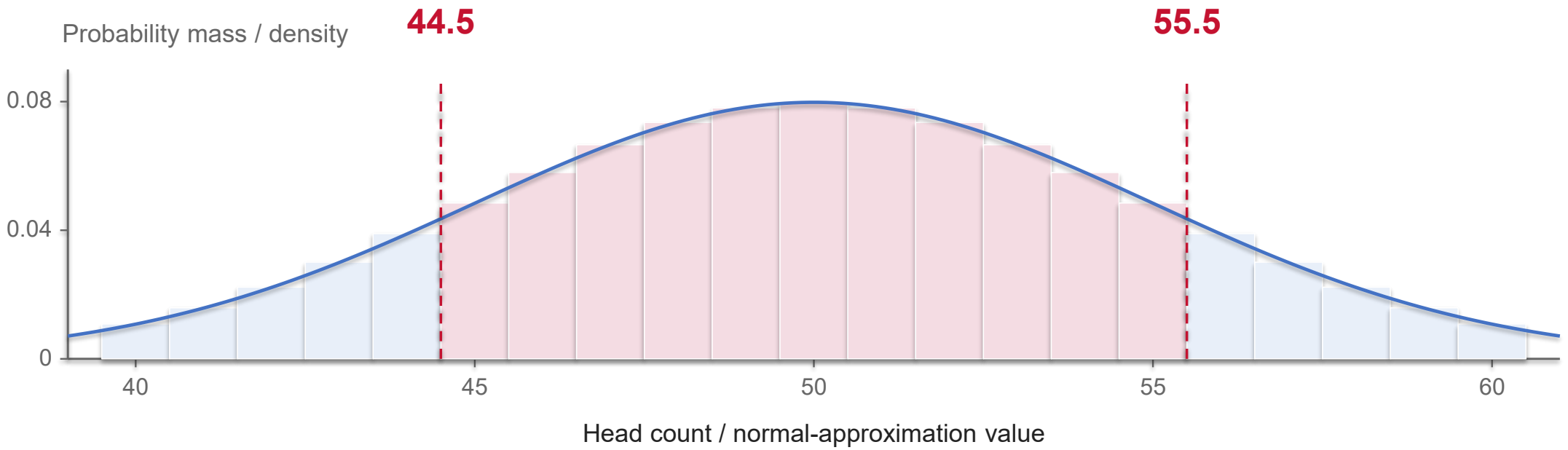
$$\mu_X = 100(0.5) = 50, \qquad \sigma_X^2 = 100(0.5)(0.5) = 25$$

$$Y \sim N(50, 25), \qquad \sigma_Y = 5$$

Both checks pass: np = 50 > 10 and n(1 − p) = 50 > 10.

Example 14: keep the full strips at 45 and 55

INCLUSIVE COUNTS 45,…,55 BECOME CONTINUOUS BOUNDARIES 44.5 AND 55.5



$$P(45 \le X \le 55) \approx P(44.5 < Y < 55.5)$$

Do not use 45 and 55 as the normal limits; that omits half of each endpoint strip.

Example 14: corrected z-scores and normal area

STANDARDIZE THE ADJUSTED COUNT LIMITS USING σ = 5

$$z_L = \frac{44.5 - 50}{5} = -1.10, \qquad z_U = \frac{55.5 - 50}{5} = 1.10$$

$$\Phi(1.10) - \Phi(-1.10) \approx 0.7287$$

About 72.87% of 100-toss outcomes fall between 45 and 55 heads.

Normal table: 0.8643 − 0.1357 = 0.7286 (rounding difference).

Correct the count limits first: 44.5 and 55.5 give z = −1.10 and +1.10.

CLT and approximation problems: avoid these traps

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FIRST IDENTIFY THE VARIABLE WHOSE PROBABILITY IS REQUESTED

Incorrect move	Correct check
“n > 30 makes my data normal.”	The CLT concerns the mean/sum distribution.
Standardize $\bar{X}$ with σ.	Use its SD, $\sigma/\sqrt{n}$.
Write a Poisson approximation $N(\lambda, \lambda^2)$.	Use variance λ and SD $\sqrt{\lambda}$.
Add the two machine standard deviations.	Add independent variances; then square-root.
Apply ±0.5 to every normal calculation.	Only use it for the integer count approximation.

A correct formula used for the wrong variable still gives a wrong probability.

Concept check: choose the rule and explain why

DISCUSS BRIEFLY BEFORE THE ANSWERS APPEAR

A If an exponential event rate doubles, what happens to the mean wait?

B After already waiting s, what enters the exponent for another t-unit wait?

C For a sum of n iid observations, is the SD σ/n, $\sigma\sqrt{n}$, or $n\sigma$?

D Why is a half-unit correction used for heads but not for machine-hours?

E For a Poisson count with mean λ, what is the normal approximation's SD?

Give the variable, the rule, and one sentence of reasoning—not just a symbol.

Concept check: answers and reasoning

CHECK THE REASONING AS WELL AS THE FINAL EXPRESSION

Item	Answer	Reason
A	Mean wait is halved.	$E[T] = 1/\lambda$, so $1/(2\lambda)$ is half as large.
B	Use $\exp(-\lambda t)$.	The elapsed s cancels after conditioning.
C	$\sigma\sqrt{n}$	$\mathrm{Var}(S_n) = n\sigma^2$ under independence.
D	Correct heads only.	Counts are discrete; machine-hours are continuous.
E	$\sqrt{\lambda}$	A Poisson count has variance λ, not λ^2.

Explain why a rule applies before inserting numbers.

Lecture 12: choose the model by the question

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THE RANDOM VARIABLE DETERMINES THE METHOD

What is being described?	Model or method	Essential check
Time to the next Poisson event	Exponential waiting time	Rate and time units agree.
An additional wait after no event	Memoryless property	Use the extra duration.
A mean or sum of many observations	CLT normal approximation	Independence; finite variance; correct SD.
A large binomial or Poisson count	Normal approximation	Check moments and count boundaries.

Distinguish exact model identities, exact moment rules, and approximate normal probabilities.

Questions and discussion

Before you calculate, ask:

Variable	Is it a count, a waiting time, a sum, or a mean?
Assumptions	What justifies this model or approximation?
Scale	Are the units, variance and standard deviation consistent?
Event	Does the boundary include an integer, and is a correction needed?

Define → check assumptions → match units → translate the event → calculate → verify.